



COMPLEX ENGINEERING PROBLEMS & DESIGN THINKING MATRICES

Process Manual Sections 1.5, 2.2, 2.5 & 2.6: Attribute Mapping, Comparative Paradigms & Risk Mitigations

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1. Addressing Complex Problems via Design Thinking (Section 2.5 Table)

Per Section 2.5 of the Process Manual, Design Thinking provides an empathetic, iterative framework that directly overcomes the inherent challenges of complex engineering problems:

Complex Problem Attribute	Process Manual Theoretical Principle	Implementation in Alliance Smart Campus Digital Twin
Ill-defined / Vague Need	Empathy & Define phases uncover real needs; reframing turns ambiguity into actionable statements.	Observations revealed facility managers did not need more raw sensor tables, but a spatial 3D overview of campus heat and power leaks.
Deep Technical Knowledge	Ideation generates technical and non-technical concepts; later phases filter for feasibility.	Coupled mathematical solar forecasting with Three.js WebGL shaders to make complex power-flow dynamics intuitively visible.
Open-Ended / Many Solutions	Divergent-convergent pattern explores wide solution space before systematic narrowing.	Evaluated desktop native apps, heavy BIM viewers, and 2D charts before converging on a zero-install WebGL 3D browser architecture.
Conflicting Constraints	Prototyping tests trade-offs (e.g., visual fidelity vs. performance) with real users.	Tested low-poly vs. high-poly campus models with facility technicians to find optimal balance between 60 FPS speed and landmark recognizability.
Requires Information Gathering	User feedback becomes a primary data source, complementing technical literature.	Combined 12 IEEE research papers with empirical interviews from Alliance electrical maintenance engineers and administrative staff.
Non-Standard Solution Path	Iteration encourages exploring unexpected paths after each user testing round.	Pivoted from static 3D building visualization to autonomous agent-driven space reclamation upon discovering 28.4% ghost bookings.
Real-World Constraints	Prototypes reveal constraints early; testing uncovers ethical or privacy concerns.	Addressed student privacy concerns by rejecting facial recognition in favor of aggregated, anonymous infrared headcount telemetry.
Requires Teamwork	Workshops, affinity mapping, and sprint reviews foster collaboration and shared mental models.	Weekly sprint syncs and modular TypeScript contracts enabled 4 student engineers to integrate graphics, power, and IoT seamlessly.

2. Design Thinking Challenges & Mitigations (Section 2.6 Table)

Process Manual Challenge	Process Manual Recommended Mitigation	Specific Project Execution Strategy
Linear Thinking Resistance	Use graded milestones for each iteration loop; demonstrate historical engineering failures.	Conducted weekly milestone reviews enforcing 3 distinct prototype evolutions (wireframe -> low-poly -> physics-coupled twin).
Time Pressure (1 Semester)	Start with low-fidelity prototypes in first 4 weeks; technical refinement later.	Delivered basic WebGL block model in Week 3, deferring procedural solar optimization and telemetry hydration to Weeks 7–9.
Finding Real Stakeholders	Partner with campus clinics, local entities, or use student surrogate role-playing.	Engaged Alliance University campus electrical maintenance technicians and ASAC lab superintendents as primary evaluators.
Empathy vs. Technical Rigor	Assign separate grades for user validation and engineering analysis; require both.	Balanced human-centered UI design (SUS score 86.4) with rigorous differential equations for thermal mass and battery degradation.
Team Conflict over Changes	Use design sprint techniques with clear decision rules.	Applied Pugh Multi-Criteria Decision Matrix with explicit weighting criteria to resolve architectural disputes objectively.